

SERMORELIN

GHRH (1-29) • Growth Hormone-Releasing Hormone Analog • Research Overview • July 2026

WHAT IT IS

Sermorelin acetate is a synthetic 29-amino-acid peptide corresponding to the first 29 amino acids of human growth hormone-releasing hormone (GHRH 1-29). This is the minimal fragment that retains full biological activity at the GHRH receptor. It was FDA-approved in 1990 (diagnostic) and 1997 (therapeutic for pediatric GH deficiency) under the brand name Geref. Commercial manufacture was discontinued in 2008 for commercial reasons (not safety). It is now available only as a compounded medication. It stimulates the pituitary to release endogenous GH in a physiological, pulsatile pattern while preserving natural somatostatin feedback.

KEY CHARACTERISTICS

FEATURE	DETAILS
Structure	Synthetic GHRH(1-29)-NH ₂ (native sequence fragment)
Half-life	~10–12 minutes (plasma); biological GH pulse lasts ~2–4 hours
Mechanism	GHRH receptor agonist → cAMP/PKA → pulsatile endogenous GH release
FDA History	Approved 1990/1997 (Geref); discontinued 2008 for commercial reasons
Typical Adult Dose	200–500 mcg SC once daily at bedtime (common clinic range)
Timing	Bedtime (fasted) to align with natural nocturnal GH pulse

KEY HUMAN EVIDENCE

Pediatric GH deficiency (Thorner et al., JCEM 1996) — multicenter trial of 110 children: 30 mcg/kg SC daily at bedtime nearly doubled height velocity from 4.1 to 8.0 cm/year at 6 months (74% responders). Well tolerated.

Adult / aging studies — Khorram 1997 (elderly, 10 mcg/kg nightly × 16 weeks): increased lean mass in men, improved skin thickness. Vittone 1997: restored more youthful GH pulsatility and raised IGF-1 in older men. Overall evidence base is older and thinner than for tesamorelin, but longer than for most research peptides.

REPORTED EFFECTS & POTENTIAL BENEFITS

- **Documented:** Increases endogenous GH secretion and IGF-1; accelerated linear growth in GH-deficient children (former indication).
- **Commonly reported (clinic/community):** Improved deep sleep quality and recovery, mild body-composition support, better skin quality, joint comfort, sense of well-being.
- **Theoretical / extrapolated:** Support for recovery, collagen, and age-related GH decline — effects are generally milder than with longer-acting GHRH analogs or exogenous GH.

COMMON RESEARCH / COMPOUNDING PROTOCOLS

Adult wellness / anti-aging range: 200–500 mcg SC once daily, preferably at bedtime on an empty stomach (aligns with the dominant nocturnal GH pulse). Many protocols use 5 nights on / 2 nights off.

Clinical / studied range: Pediatric label was 30 mcg/kg/day. Adult trials used roughly 0.5–10 mcg/kg or fixed doses around 1–2 mg in older studies.

Administration: Subcutaneous (abdomen preferred, rotate sites). Reconstitute lyophilized powder with bacteriostatic water; refrigerate and use within ~28 days. Typical reconstitution example: 3 mg + 3 mL BAC → 1,000 mcg/mL (300 mcg = 30 units on a U-100 syringe).

SAFETY PROFILE & CONSIDERATIONS

Sermorelin has one of the better-characterized safety profiles among GHRH analogs because of its prior FDA approval and multi-year pediatric use. **Most common effects:** injection-site reactions (redness, pain, swelling — typically mild), transient facial flushing, occasional headache or dizziness. Metabolic effects (mild reductions in insulin sensitivity, transient lipid changes) have been observed in some adult studies. Because it is short-acting and feedback-regulated, the risk of sustained supraphysiologic GH/IGF-1 is lower than with long-acting analogs or exogenous GH. Still not approved for adult anti-aging use. Avoid with active malignancy. Monitor IGF-1 and glucose if used under medical supervision.

RELATION TO CJC-1295, TESAMORELIN & OTHER GHRH ANALOGS

Sermorelin is the native-sequence GHRH(1-29) fragment and the historical reference compound. **CJC-1295 (no DAC / Mod GRF)** is a stabilized version of the same 1-29 backbone with four amino-acid substitutions for longer half-life (~30 min). **CJC-1295 with DAC** extends action to days via albumin binding. **Tesamorelin** is a full-length (1-44) GHRH analog with an N-terminal modification; it is the only currently FDA-approved GHRH peptide (HIV lipodystrophy) and has the strongest modern clinical data, especially for visceral fat. Sermorelin is often viewed as the gentlest / most physiological option among the three.

DISCLAIMER: For educational and research reference only. Sermorelin is no longer commercially manufactured as an approved drug product and is available only via compounding. It is not approved for adult anti-aging or body-composition use. Use outside supervised medical settings carries risks. Consult a qualified physician before considering any peptide protocol. Information synthesized from published literature (Thorner 1996, Khorram 1997, etc.) and public peptide research summaries as of July 2026.